CHEAT SHEET 1 • SIDE A

THE SECOND HALF: ESTIMATION AND ASYMPTOTICS

ordered by exam probability, highest first

Formulas and procedural steps only. θ_0 true value (fixed) • θ any candidate • $\widehat{\theta}$ estimator (random) • $\overline{X} = \frac{1}{n}\sum X_i$ • $S^2 = \frac{1}{n-1}\sum(X_i - \overline{X})^2$ • $\widetilde{S}^2 = \frac{1}{n}\sum(X_i - \overline{X})^2 = \frac{n-1}{n}S^2$

0. WHICH MACHINE DOES THIS QUESTION WANT?

Trigger phrase	Go to
"find the MOM estimator"	box 3
"find the MLE"	box 2
"find the CRLB"	box 1, steps 0–3
"is it a UMVUE?"	box 5
"limiting distribution of"	box 1, step 3
"asymptotic / approximate dist."	box 1, step 4
"how would you estimate its variance"	box 1, step 4
"estimate $g(\theta_0)$ and its distribution"	box 4
"show it is consistent"	box 6
"which has lower MSE"	box 10
"consistent, asy. normal, asy. efficient"	the MLE, box 11
"construct a χ^2 / t / F"	box 8
"prove $E(\overline{X})$, $\mathrm{Var}(\overline{X})$, $E(S^2)$"	box 9
"verify the score / info identity"	box 12
two sequences, one $\xrightarrow{d}$ one $\xrightarrow{p}$	Slutsky, box 11
one estimator through a function	delta, box 4
support of x contains θ	box 13 first

1. THE MASTER PIPELINE (answers four question types at once)

0. $\ln f(x \mid \theta)$ ONE observation, expand fully

1. $S(\theta, X) = \dfrac{\partial \ln f(X \mid \theta)}{\partial \theta}$ *no summation sign*

2. $I(\theta_0) = -E\left[\dfrac{\partial S(\theta_0, X)}{\partial \theta}\right] = \mathrm{Var}[S(\theta_0, X)] = E[S^2(\theta_0, X)]$

3. CRLB $= \dfrac{1}{nI(\theta_0)}$ and $\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N\left(0, [I(\theta_0)]^{-1}\right)$

4. $\widehat{\theta}_{MLE} \overset{a}{\sim} N\left(\theta_0, \dfrac{1}{nI(\theta_0)}\right)$; $\widehat{\mathrm{AVar}} = \dfrac{1}{nI(\theta)}\Big|_{\theta=\widehat{\theta}}$

Choosing the $I(\theta_0)$ formula. Score linear in X ⇒ $\mathrm{Var}[S]$, one line. Score nonlinear ⇒ differentiate again, $-E[\partial S/\partial\theta]$. Regularity fails ⇒ use the definition $E[S^2]$.

Key fact: AVar of the MLE = CRLB. That is asymptotic efficiency.

$I(\theta_0)$ is per observation. If an n appears inside your $I(\theta_0)$, you summed the score by mistake.

2. MLE STEPS

1. Log the density for one observation, expand fully.

2. $L(\theta) = \sum_{i=1}^{n} \ln f(X_i \mid \theta)$; terms free of θ become constants.

3. $\partial L/\partial\theta = 0$ (first-order condition).

4. Solve for θ; write $\widehat{\theta}_{MLE}$ with capital X's.

- Second-order conditions are not required.
- Parameter σ^2: differentiate with respect to σ^2, not σ. Set $v = \sigma^2$ and work in v.
- Support depends on θ: calculus fails. Impose $\theta \geq \max\{X_i\}$, then use monotonicity of $-n\ln\theta$.
- Absolute value in the density: not differentiable; the MLE is the sample median.
- No closed form (two-parameter Gamma): say numerical maximization is required, and use MOM instead.

Invariance. MLE of $g(\theta_0)$ is $g(\widehat{\theta}_{MLE})$. Plug in, no new work. Used for $\sqrt{\theta}$, $1/\theta$, θ^2, $e^{-\theta}$, and for $\widehat{\mathrm{AVar}}$.

3. MOM STEPS

1. Count parameters k; you need k equations.

2. Write population moments as functions of θ; integrate if not given.

3. Set equal to sample counterparts, *centered*:

$$E(X) = \overline{X}, \qquad \mathrm{Var}(X) = \widetilde{S}^2$$

4. Solve the system; label $\widehat{\theta}_{MOM}$.

- NEVER pair $\mathrm{Var}(X)$ with $\frac{1}{n}\sum X_i^2$, nor $\widetilde{S}^2$ with $E(X^2)$.
- Uncentered alternative: $E(X) = \overline{X}$ and $E(X^2) = \frac{1}{n}\sum X_i^2$.
- The sample second moment uses $\frac{1}{n}$, not $\frac{1}{n-1}$.
- If both equations are products of powers of the parameters, divide one by the other to eliminate one parameter.
- If the parameter is a standard deviation, take a square root at the end.
- Drawbacks: can fall outside Θ, may be inefficient. Virtue: almost always closed form.

4. DELTA METHOD STEPS (estimating $\gamma_0 = g(\theta_0)$)

1. Estimator by invariance: $\widehat{\gamma} = g(\widehat{\theta})$.

2. Compute $g'(\theta)$, square it, evaluate at θ_0.

3. Multiply by Λ from $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, \Lambda)$. For an MLE, $\Lambda = [I(\theta_0)]^{-1}$.

4. Write both forms:

$$\sqrt{n}\left[g(\widehat{\theta}) - g(\theta_0)\right] \xrightarrow{d} N\left(0, [g'(\theta_0)]^2\Lambda\right)$$

$$g(\widehat{\theta}) \overset{a}{\sim} N\left(g(\theta_0),\ \tfrac{1}{n}[g'(\theta_0)]^2\Lambda\right)$$

Conditions: g continuously differentiable, $g'(\theta_0) \neq 0$. Center at $g(\theta_0)$, never at θ_0. Square the derivative; it enters the variance slot.

Squared derivatives you may need:

$g = \sqrt{\theta} \Rightarrow [g']^2 = \frac{1}{4\theta}$ $\quad$ $g = 1/\theta \Rightarrow [g']^2 = \theta^{-4}$

$g = \theta^2 \Rightarrow [g']^2 = 4\theta^2$ $\quad$ $g = e^{-\theta} \Rightarrow [g']^2 = e^{-2\theta}$

$g = \ln\theta \Rightarrow [g']^2 = \theta^{-2}$ $\quad$ $g = \theta^k \Rightarrow [g']^2 = k^2\theta^{2k-2}$

5. UMVUE DECISION STEPS

1. Unbiased? If $E(\widehat{\theta}) \neq \theta_0$, STOP. Answer "No: biased estimators cannot be UMVUE." Nothing further is needed.

2. Compute $\mathrm{Var}(\widehat{\theta})$ and compare with CRLB $= 1/(nI(\theta_0))$.

3. Unbiased AND Var = CRLB ⇒ it is the UMVUE, and no unbiased estimator has smaller variance.

- Converse is false: a UMVUE need not attain the bound, so Var > CRLB does not prove "not UMVUE."
- The CRLB bounds the VARIANCE of UNBIASED estimators only. A biased estimator may sit below it legitimately.
- Definition: (i) $E(\widehat{\theta}^*) = \theta_0$ for all $\theta_0 \in \Theta$; (ii) $\mathrm{Var}(\widehat{\theta}^*) \leq \mathrm{Var}(\widehat{\theta})$ for every other unbiased $\widehat{\theta}$ and every θ_0. "Uniform" means at all θ_0.
- Restricting to unbiased estimators is what rules out constant estimators.

6. CONSISTENCY PROOF STEPS

1. Compute $E(\widehat{\theta})$, hence Bias $= E(\widehat{\theta}) - \theta_0$; show $\lim_{n\to\infty}$ Bias $= 0$.

2. Compute $\mathrm{Var}(\widehat{\theta})$; show $\lim_{n\to\infty}\mathrm{Var}(\widehat{\theta}) = 0$.

3. Conclude in these words: both limits are zero, so $\widehat{\theta} \xrightarrow{q.m.} \theta_0$; mean-square convergence implies convergence in probability, so $\widehat{\theta} \xrightarrow{p} \theta_0$; therefore $\widehat{\theta}$ is consistent for θ_0.

Shortcuts. WLLN gives $\overline{X} \xrightarrow{p} \mu$ in one line. Continuous mapping: $\widehat{\theta} \xrightarrow{p} \theta_0$ and g continuous ⇒ $g(\widehat{\theta}) \xrightarrow{p} g(\theta_0)$, i.e. $\mathrm{plim}\, g(\widehat{\theta}) = g(\mathrm{plim}\,\widehat{\theta})$.

Definition: $\lim_{n\to\infty} P(|\widehat{\theta} - \theta_0| > \epsilon) = 0$ for every $\epsilon > 0$.

Biased and consistent are compatible; asymptotic unbiasedness suffices.

Converse fails: consistency does not require the variance to go to zero.

7. VARIANCE AND EXPECTATION EXTRACTION

$E(a + bY) = a + bE(Y)$, $\quad \mathrm{Var}(a + bY) = b^2\mathrm{Var}(Y)$

$\mathrm{Var}(Y) = E(Y^2) - [E(Y)]^2 \quad \Leftrightarrow \quad E(Y^2) = \mathrm{Var}(Y) + [E(Y)]^2$

If $cY \sim \chi^2_{(r)}$: $E(Y) = r/c$ and $\mathrm{Var}(Y) = 2r/c^2$

$\mathrm{Var}(\sum_{i=1}^{n} Y_i) = \sum_i \mathrm{Var}(Y_i)$ *only under independence*

$\mathrm{Var}(aX + bY) = a^2\mathrm{Var}X + b^2\mathrm{Var}Y + 2ab\mathrm{Cov}(X, Y)$

$\mathrm{Var}(\overline{X}) = \mathrm{Var}(X)/n$; $\mathrm{Var}(c\overline{X}) = c^2\mathrm{Var}(X)/n$

If $\widehat{\theta} = c\overline{X}$: $E(\widehat{\theta}) = c\,E(X)$, $\mathrm{Var}(\widehat{\theta}) = c^2\mathrm{Var}(X)/n$

If $\widehat{\theta} = \frac{1}{cn}\sum h(X_i)$: $E(\widehat{\theta}) = \frac{E[h(X)]}{c}$, $\mathrm{Var}(\widehat{\theta}) = \frac{\mathrm{Var}[h(X)]}{c^2 n}$

8. PROOF SKELETONS (annotate every line; the annotations are the grade)

$E(\overline{X}) = \mu$**.** definition of $\overline{X}$ → linearity, pull out $\frac{1}{n}$ → expectation of a sum is the sum of expectations → *identically distributed*, so each $E(X_i) = \mu$ → $\frac{1}{n}\cdot n\mu$.

$\mathrm{Var}(\overline{X}) = \sigma^2/n$**.** definition → $\mathrm{Var}(aY) = a^2\mathrm{Var}(Y)$, so $\frac{1}{n}$ becomes $\frac{1}{n^2}$ → **by independence**, $\mathrm{Cov}(X_i, X_j) = 0$, so the variance of the sum is the sum of variances → identically distributed → $\frac{1}{n^2}n\sigma^2$.

$E(S^2) = \sigma^2$**.** start from $(n-1)S^2 = \sum X_i^2 - n\overline{X}^2$ → take expectations → apply the flip to both pieces, $E(X_i^2) = \sigma^2 + \mu^2$ and $E(\overline{X}^2) = \frac{\sigma^2}{n} + \mu^2$ → the μ^2 terms cancel and $n-1$ copies of σ^2 remain.

Two independent samples. same two chains, plus one extra sentence killing $\mathrm{Cov}(\overline{X}, \overline{Y}) = 0$ across samples before splitting the variance.

Named justifications to write: linearity of expectation; $\mathrm{Var}(aY) = a^2\mathrm{Var}(Y)$; by independence; identically distributed; Thm 5.2.6; Thm 5.3.1(c).

8. BUILD A χ^2, t, OR F

1. Count what you need. $t_{(p)}$: one $N(0,1)$ plus an independent $\chi^2_{(p)}$. $F_{(r_1,r_2)}$: two independent chi-squares.

2. Standardize every normal: $Z_j = \dfrac{X_j - \mu_j}{\sigma_j} \sim N(0,1)$.

Divide by the standard *deviation*.

3. Build chi-squares by squaring and summing as many Z_j as you need degrees of freedom.

4. Form the ratio, dividing each χ^2 by its OWN df, and state which variables are independent of which.

Never reuse a variable in numerator and denominator: independence would fail. Allocate one variable to the top, the rest to the bottom.

Building blocks.

$Z \sim N(0,1) \Rightarrow Z^2 \sim \chi^2_{(1)}$

$Y_i \sim \chi^2_{(p_i)}$ independent $\Rightarrow \sum Y_i \sim \chi^2_{(\sum p_i)}$

m independent $N(0,1)$, squared and summed $\Rightarrow \chi^2_{(m)}$

$U \sim N(0,1)$, $V \sim \chi^2_{(p)}$, independent:

$$\frac{U}{\sqrt{V/p}} \sim t_{(p)}$$

$U \sim \chi^2_{(r_1)}$, $V \sim \chi^2_{(r_2)}$, independent:

$$\frac{U/r_1}{V/r_2} \sim F_{(r_1,r_2)}$$

Shapes: $t = \dfrac{\text{normal}}{\sqrt{\chi^2/\text{its own df}}}$, $\quad F = \dfrac{\chi^2/\text{its own df}}{\chi^2/\text{its own df}}$

The t has a square root over the whole denominator; the F does not.

To simplify a messy ratio, regroup so each squared deviation sits over its own variance.

9. SAMPLING DISTRIBUTION FACTS

ANY f_X with mean μ, variance σ^2 (Thm 5.2.6). No normality needed.

$$E(\overline{X}) = \mu, \quad \text{Var}(\overline{X}) = \frac{\sigma^2}{n}, \quad E(S^2) = \sigma^2$$

NORMAL ONLY (Thm 5.3.1).

$$\overline{X} \sim N\left(\mu, \tfrac{\sigma^2}{n}\right); \quad \overline{X} \perp S^2; \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

Consequences under normality.

$$\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \sim N(0,1), \quad \frac{\sqrt{n}(\overline{X}-\mu)}{S} \sim t_{(n-1)}$$

$$\frac{\sum(X_i-\mu)^2}{\sigma^2} \sim \chi^2_{(n)}, \quad \frac{n(\overline{X}-\mu)^2}{\sigma^2} \sim \chi^2_{(1)}$$

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1}, \qquad \text{Var}(\widetilde{S}^2) = \frac{2(n-1)\sigma^4}{n^2}$$

$$E(\widetilde{S}^2) = \frac{n-1}{n}\sigma^2, \qquad \text{Bias}(\widetilde{S}^2) = -\frac{\sigma^2}{n}$$

Mean known to be 0: $\frac{1}{n}\sum X_i^2$ is unbiased for σ^2, with variance $\frac{2\sigma^4}{n}$.

Degrees of freedom rule: subtracting the known μ gives n; subtracting the estimated $\overline{X}$ costs one and gives $n-1$.

9b. TWO SAMPLES, AND FOUR IDENTITIES

$$E(\overline{X}-\overline{Y}) = \mu_x - \mu_y, \quad \text{Var}(\overline{X}-\overline{Y}) = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}$$

(plus sign: independent variances add)

$$\frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2} \sim F_{(n-1,m-1)}$$

$\text{Cov}(X_i - \overline{X}, \overline{X}) = 0$; $\sum(X_i - \overline{X}) = 0$;

$(n-1)S^2 = \sum X_i^2 - n\overline{X}^2$; $M_{\overline{X}}(t) = [M_X(t/n)]^n$

10. MSE AND BIAS

$$\text{MSE}(\widehat{\theta}) = E[(\widehat{\theta}-\theta_0)^2] = \text{Var}(\widehat{\theta}) + \text{Bias}^2(\widehat{\theta})$$

$$\text{Bias}(\widehat{\theta}) = E(\widehat{\theta}) - \theta_0$$

Unbiased $\Rightarrow$ MSE = Var: half of all MSE questions are variance questions.

Derivation if asked: put $Y = \widehat{\theta} - \theta_0$ into $E(Y^2) = \text{Var}Y + [EY]^2$, and note $\text{Var}(\widehat{\theta} - \theta_0) = \text{Var}(\widehat{\theta})$ since θ_0 is a constant.

Comparison steps: for each estimator get E, then bias, then Var, then add, then compare. **If the ranking depends on n or on θ_0, state the condition explicitly.**

Variance shrinks with n; a fixed bias does not, so a non-vanishing bias eventually dominates.

11. ASYMPTOTIC THEOREMS

Chain: $\xrightarrow{q.m.} \Rightarrow \xrightarrow{p} \Rightarrow \xrightarrow{d}$. No reverses.

$X_n \xrightarrow{q.m.} c$: $\lim E[(X_n - c)^2] = 0$

$X_n \xrightarrow{p} c$: $\lim P(|X_n - c| > \epsilon) = 0$ for all $\epsilon > 0$

$X_n \xrightarrow{d} Y$: $\lim F_{X_n}(r) = F_Y(r)$

WLLN: $\overline{X} \xrightarrow{p} \mu$. *Proof:* Chebyshev gives $P(|\overline{X} - \mu| > \epsilon) \le \frac{\text{Var}(\overline{X})}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \to 0$.

CLT (Lindeberg-Lévy): for any f_X with mean μ and variance σ^2,

$$\sqrt{n}(\overline{X}-\mu) \xrightarrow{d} N(0,\sigma^2), \quad \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \xrightarrow{d} N(0,1)$$

$$\Longrightarrow \quad \overline{X} \overset{a}{\sim} N\left(\mu, \tfrac{\sigma^2}{n}\right)$$

The CLT is about $\overline{X}$, not the individual X_i. The $\sqrt{n}$ is the unique rate that stabilizes the variance.

11b. SLUTSKY, EFFICIENCY, THE MLE THEOREM

Probability limit algebra: $X_n \xrightarrow{p} a$, $Y_n \xrightarrow{p} b \Rightarrow X_n + Y_n \xrightarrow{p} a+b$; $X_n Y_n \xrightarrow{p} ab$; $X_n/Y_n \xrightarrow{p} a/b$ if $b \neq 0$.

Slutsky: $X_n \xrightarrow{d} U$ (random), $Y_n \xrightarrow{p} a$ (constant):

$$X_n Y_n \xrightarrow{d} aU, \quad X_n + Y_n \xrightarrow{d} U + a, \quad \frac{X_n}{Y_n} \xrightarrow{d} \frac{U}{a} \ (a \neq 0)$$

Pulling a constant into a normal squares it: $a \cdot N(0,V) = N(0, a^2 V)$.

Asymptotic efficiency: $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$, i.e. asymptotically unbiased with AVar = CRLB.

MLE theorem. $\widehat{\theta}_{MLE}$ is (a) consistent, (b) asymptotically normal, $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, [I(\theta_0)]^{-1})$, (c) asymptotically efficient, $\widehat{\theta} \overset{a}{\sim} N(\theta_0, [nI(\theta_0)]^{-1})$.

The MLE is usually *biased* in finite samples.

12. THE TWO IDENTITIES

Score identity: $E[S(\theta_0, X)] = 0$

Information matrix identity:

$$-E\left[\frac{\partial S(\theta_0, X)}{\partial \theta}\right] = E\left[S^2(\theta_0, X)\right]$$

Both hold ONLY at θ_0. At a general θ they fail, and that failure is usually the point of the question.

Since $E[S] = 0$, $\text{Var}[S] = E[S^2]$, which is why two of the three $I(\theta_0)$ formulas coincide.

$E[(X-\mu)^2]$ equals $\text{Var}(X)$ only at the true μ.

Verification steps: substitute θ_0, take the expectation, quote the true $E(X)$, $E(X^2)$ or $E(X^4)$, and show it collapses to 0 or to equality.

Origin: differentiate $\int f(x \mid \theta)\,dx = 1$ once (score) or twice (information) with respect to θ.

13. WHEN THE THEOREMS FAIL

Detector: does the range of x contain θ? Tells: support $(0,\theta_0)$, Uniform$(0,\theta_0)$, an indicator $\mathbf{1}\{x \le \theta\}$, or a score with no X in it.

Then differentiation and integration cannot be interchanged, and:

- the CRLB need not bind, so an unbiased estimator can have variance strictly below it;
- the score identity can fail, $E[S] \neq 0$;
- the three $I(\theta_0)$ formulas can disagree, so use the definition $1/(nE[S^2])$;
- the MLE cannot be found by setting a derivative to zero.

Write the sentence: "the support depends on θ_0, so the regularity conditions fail and the theorem does not apply."

14. NICHE BUT USEFUL

$E(\chi^2_{(r)}) = r$, $\text{Var}(\chi^2_{(r)}) = 2r$

$X \sim F_{(p,q)} \Rightarrow 1/X \sim F_{(q,p)}$

$X \sim t_{(q)} \Rightarrow X^2 \sim F_{(1,q)}$, hence $\frac{n(\overline{X}-\mu)^2}{S^2} \sim F_{(1,n-1)}$

$X \sim N(0,\sigma^2)$: $E(X^2) = \sigma^2$, $E(X^4) = 3\sigma^4$, odd moments zero.

$E(S^2) = \sigma^2$ does NOT give $E(S) = \sigma$; by Jensen, $E(S) < \sigma$.

$E[g(\widehat{\theta})] \neq g[E(\widehat{\theta})]$, but $\text{plim}\, g(\widehat{\theta}) = g(\text{plim}\, \widehat{\theta})$.

Exponential: $\frac{1}{\theta}e^{-x/\theta}$ has $E = \theta$ and MLE $\overline{X}$; $\theta e^{-\theta x}$ has $E = 1/\theta$ and MLE $1/\overline{X}$. Read which one you are given.

Geometric $p(1-p)^{x-1}$: $E(X) = 1/p$, so $\overline{X}$ is biased for p and $\widetilde{p}_{MOM} = 1/\overline{X}$.

$\widehat{\theta}_{MOM} = \widehat{\theta}_{MLE}$ for Normal, Poisson, Bernoulli and exponential-in-mean; not in general.

Random sample = independent + identically distributed; joint density $\prod_i f(x_i)$. A statistic is any function of the sample, hence random.

GMM: more moment conditions than parameters, model over-identified, minimize the distance between sample and population moments.

CHEAT SHEET 2 • SIDE A

THE PROBABILITY TOOLKIT (trimmed midterm sheet)

the first-half tools the final actually uses

Everything here is a tool the estimation questions call on. Set theory, Kolmogorov's axioms, Bayes' rule and the joint-distribution machinery are compressed to Side B, since they are unlikely to carry a question of their own.

1. EXPECTATION AND VARIANCE ALGEBRA (used in almost every answe

$E(aX+b) = aE(X)+b$
$E(X+Y) = E(X)+E(Y)$ *always, no independence needed*
$E(aX+bY) = aE(X)+bE(Y)$
$E(c) = c$ and $\mathrm{Var}(c) = 0$ for a constant c
$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ *shifts do not change variance*
$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$
$\boxed{E(X^2) = \mathrm{Var}(X) + [E(X)]^2}$ *the flip; use it whenever an* $E(\cdot^2)$ *blocks you*
$\mathrm{Var}(aX+bY) = a^2\mathrm{Var}X + b^2\mathrm{Var}Y + 2ab\mathrm{Cov}(X,Y)$
$\mathrm{Var}(X-Y) = \mathrm{Var}X + \mathrm{Var}Y - 2\mathrm{Cov}(X,Y)$
$\mathrm{Var}(\sum_i X_i) = \sum_i \mathrm{Var}(X_i)$ *iff all covariances vanish*
$E[g(X)] \neq g[E(X)]$ in general.

2. COMPUTING MOMENTS FROM A DENSITY

$$E(X) = \int_{\text{supp}} x f(x)\,dx, \quad E(X^2) = \int_{\text{supp}} x^2 f(x)\,dx$$

$$E[g(X)] = \int_{\text{supp}} g(x) f(x)\,dx \quad \text{(LOTUS)}$$

Discrete: replace $\int$ by $\sum_x$ and f by the pmf.
Always integrate over the stated support, not over $\mathbb{R}$.
Sanity check: $\int_{\text{supp}} f(x)\,dx = 1$.

Integrals that keep appearing:
$\int_0^\theta x^k dx = \dfrac{\theta^{k+1}}{k+1}$ $\quad$ $\int_0^1 x^k dx = \dfrac{1}{k+1}$
$\int_0^\infty x^k e^{-\lambda x} dx = \dfrac{k!}{\lambda^{k+1}}$ (integer $k \geq 0$)
$\int_0^\infty x^{\alpha-1} e^{-x/\beta} dx = \Gamma(\alpha)\beta^\alpha$
An odd function integrated over a symmetric interval gives zero.

3. COVARIANCE AND CORRELATION

$\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$
$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; $\mathrm{Cov}(X,c) = 0$
$\mathrm{Cov}(aX+b,\ cY+d) = ac\,\mathrm{Cov}(X,Y)$
$\mathrm{Cov}(X+W, Y+Z) = \mathrm{Cov}(X,Y) + \mathrm{Cov}(X,Z) + \mathrm{Cov}(W,Y) + \mathrm{Cov}(W,Z)$
$\mathrm{Corr}(X,Y) = \rho = \dfrac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1,1]$
Independence $\Rightarrow$ Cov $= 0$ and $E(XY) = E(X)E(Y)$. The converse is false.

4. TRANSFORMATIONS OF RANDOM VARIABLES

CDF method (always works).
1. $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$
2. Rearrange to isolate X; flip the inequality if you divide by a negative.
3. Write it as $F_X(\cdot)$, then differentiate to get $f_Y(y)$.

Change of variable (monotone g).

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{d}{dy} g^{-1}(y)\right|$$

Find the new support by pushing the endpoints of the old support through g.

Bivariate. $f_{UV}(u,v) = f_{XY}\big(x(u,v), y(u,v)\big)\,|J|$ with

$$J = \det\begin{pmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{pmatrix}$$

then integrate out to get a marginal.

Normal facts. $X \sim N(\mu,\sigma^2) \Rightarrow \frac{X-\mu}{\sigma} \sim N(0,1)$; $aX+b \sim N(a\mu+b, a^2\sigma^2)$; sums of independent normals are normal with means and variances adding.

5. MOMENT GENERATING FUNCTIONS

$M_X(t) = E(e^{tX})$
$E(X^k) = M_X^{(k)}(0)$, the kth derivative at zero
$M_{aX+b}(t) = e^{bt} M_X(at)$
$X \perp Y \Rightarrow M_{X+Y}(t) = M_X(t) M_Y(t)$
Uniqueness: the same mgf means the same distribution. This is how you identify a distribution from its mgf.
$M_{\overline{X}}(t) = [M_X(t/n)]^n$
$N(\mu,\sigma^2)$: $M(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$

6. JOINT, MARGINAL, CONDITIONAL

Marginal: $f_X(x) = \int f_{XY}(x,y)\,dy$
Conditional: $f_{Y|X}(y \mid x) = \dfrac{f_{XY}(x,y)}{f_X(x)}$

Independence test (both parts required).
(i) $f_{XY}(x,y) = f_X(x) f_Y(y)$, and
(ii) the support is a **rectangle**: the range of y must not depend on x.
Factorization alone is not sufficient.

$E(Y \mid X = x)$ is a *number*; $E(Y \mid X)$ is a *random variable*.
Law of iterated expectations: $E[E(Y \mid X)] = E(Y)$
Variance decomposition: $\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid X)] + \mathrm{Var}[E(Y \mid X)]$
$E[g(X)Y \mid X] = g(X)\,E[Y \mid X]$ (pull out what is known)

7. THE THREE INEQUALITIES

Markov: $X \geq 0 \Rightarrow P(X \geq a) \leq \dfrac{E(X)}{a}$

Chebyshev: $P(|X-\mu| \geq k\sigma) \leq \dfrac{1}{k^2}$, equivalently

$$P(|X-\mu| > \epsilon) \leq \frac{\mathrm{Var}(X)}{\epsilon^2}$$

This is the one that proves the WLLN.
Jensen: g convex $\Rightarrow E[g(X)] \geq g[E(X)]$; g concave $\Rightarrow$ reverse. Square root is concave, hence $E(S) < \sigma$.

8. ORDER STATISTICS (needed whenever the MLE is $\max\{X_i\}$)

Let $X_{(n)} = \max\{X_i\}$ and $X_{(1)} = \min\{X_i\}$ from an i.i.d. sample.

$$F_{X_{(n)}}(x) = [F_X(x)]^n, \qquad f_{X_{(n)}}(x) = n\,[F_X(x)]^{n-1} f_X(x)$$

$$F_{X_{(1)}}(x) = 1 - [1 - F_X(x)]^n$$

$$f_{X_{(1)}}(x) = n\,[1 - F_X(x)]^{n-1} f_X(x)$$

Logic: the max is below x exactly when *all* n observations are, and independence turns that into a product. The min exceeds x exactly when all n do.
For Uniform$(0,\theta)$, $F_X(x) = x/\theta$ on $(0,\theta)$, so

$$f_{X_{(n)}}(x) = \frac{n x^{n-1}}{\theta^n}, \quad E\big[X_{(n)}\big] = \frac{n}{n+1}\theta$$

so $\max\{X_i\}$ is biased downward, and $\frac{n+1}{n}\max\{X_i\}$ corrects it.

9. READING A DENSITY (do this before anything else)

1. Find the support. The phrase "and zero otherwise" matters. Every integral runs over the support only.
2. Check whether the support involves θ. If it does, flag it now: the CRLB, the score identity and calculus-based MLE all become suspect (Sheet 1, box 13).
3. Identify the parameter and its role. Is θ the mean, the reciprocal of the mean, a variance, or a standard deviation? This decides whether the answer is $\overline{X}$, $1/\overline{X}$, $\widetilde{S}^2$ or $\widetilde{S}$.
4. Separate the normalizing constant from the part that depends on x; the constant often carries all of the θ dependence.
5. Verify $\int f = 1$ if you are unsure you copied it correctly.

10. SUMS AND CLOSURE PROPERTIES

For independent X and Y, use mgfs: $M_{X+Y}(t) = M_X(t)M_Y(t)$, then identify by uniqueness.
Closed under addition (independent case):
$N(\mu_1,\sigma_1^2) + N(\mu_2,\sigma_2^2) = N(\mu_1+\mu_2,\ \sigma_1^2+\sigma_2^2)$
Poisson(λ_1)+Poisson(λ_2) =Poisson$(\lambda_1+\lambda_2)$
$\chi^2_{(r_1)} + \chi^2_{(r_2)} = \chi^2_{(r_1+r_2)}$
Gamma(α_1,β)+Gamma(α_2,β) =Gamma$(\alpha_1+\alpha_2,\beta)$
Binomial(n_1,p)+Binomial(n_2,p) =Binomial(n_1+n_2,p)
Sum of n independent Exponential(mean θ) is Gamma(n,θ).
Not closed: Uniform, t, F.

CHEAT SHEET 2 • SIDE B

DISTRIBUTIONS, ALGEBRA, AND LEFTOVERS

lowest priority: he supplies pdfs, $E(X)$ and $\mathrm{Var}(X)$ on the exam

Per his email, any pdf, mean or variance you need will be given to you. This page is insurance and a recognition aid, not something to spend exam time on. If you run out of room while transcribing, cut here first.

8. DISTRIBUTION TABLE

Name	pdf / pmf	$E(X)$	$\mathrm{Var}(X)$	$I(\theta_0)$
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	$p(1-p)$	$\frac{1}{p(1-p)}$
Binomial(n,p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	—
Geometric(p)	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	—
Poisson(λ)	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ	$\frac{1}{\lambda}$
Uniform(a,b)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	—
Uniform$(0,\theta)$	$\frac{1}{\theta}$	$\frac{\theta}{2}$	$\frac{\theta^2}{12}$	*fails*
Expon., mean θ	$\frac{1}{\theta}e^{-x/\theta}$	θ	θ^2	$\frac{1}{\theta^2}$
Expon., rate λ	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{1}{\lambda^2}$
Gamma(α,β)	$\frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}$	$\alpha\beta$	$\alpha\beta^2$	—
$N(\mu,\sigma^2)$, σ^2 known	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\frac{1}{\sigma^2}$
$N(0,\sigma^2)$, for σ^2	as above with $\mu=0$	0	σ^2	$\frac{1}{2\sigma^4}$
Beta(α,β)	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	—
$\chi^2_{(r)}$	Gamma$(r/2,2)$	r	$2r$	—
$t_{(p)}$	ratio, Sheet 1 box 8	0 $(p>1)$	$\frac{p}{p-2}$ $(p>2)$	—
$F_{(p,q)}$	ratio, Sheet 1 box 8	$\frac{q}{q-2}$ $(q>2)$	—	—
$\theta x^{\theta-1}$, $(0,1)$	as written	$\frac{\theta}{\theta+1}$	—	$\frac{1}{\theta^2}$

Relationships. $\chi^2_{(r)} = \text{Gamma}(r/2,2)$. Exponential $=$ Gamma$(1,\beta)$. A squared $N(0,1)$ is $\chi^2_{(1)}$. A sum of r independent $\chi^2_{(1)}$ is $\chi^2_{(r)}$. A sum of independent normals is normal.

9. LOG, EXPONENT AND DERIVATIVE RULES (for MLE work)

$\ln(ab) = \ln a + \ln b$; $\ln(a/b) = \ln a - \ln b$; $\ln(a^b) = b\ln a$
$\ln(e^u) = u$; $\ln 1 = 0$; $\ln(1/a) = -\ln a$

$$\frac{d}{d\theta}\ln\theta = \frac{1}{\theta}; \qquad \frac{d}{d\theta}\ln(1-\theta) = \frac{-1}{1-\theta}$$
$$\frac{d}{d\theta}\theta^{-1} = -\theta^{-2}; \qquad \frac{d}{d\theta}\theta^{-2} = -2\theta^{-3}$$
$$\frac{d}{d\theta}e^{-\theta} = -e^{-\theta}; \qquad \frac{d}{d\theta}\sqrt{\theta} = \frac{1}{2\sqrt{\theta}}$$
$$\frac{d}{d\theta}\theta^2 = 2\theta; \qquad \frac{d}{d\theta}\frac{1}{\theta^2} = -\frac{2}{\theta^3}$$

When the parameter is σ^2, set $v=\sigma^2$ and differentiate in v:

$$\frac{d}{dv}\ln v = \frac{1}{v}, \quad \frac{d}{dv}\frac{1}{2v} = -\frac{1}{2v^2}, \quad \frac{d}{dv}\frac{1}{2v^2} = -\frac{1}{v^3}$$

10. SUMMATION FACTS

$\sum_{i=1}^n X_i = n\overline{X}$; $\sum_{i=1}^n (X_i - \overline{X}) = 0$
$\sum_{i=1}^n c = nc$ for a constant c
$\sum(X_i-\overline{X})^2 = \sum X_i^2 - n\overline{X}^2$
Pull constants out of sums before differentiating; terms without θ vanish under $\partial/\partial\theta$.

11. SET THEORY AND BASIC PROBABILITY (compressed)

$P(A) \geq 0$; $P(S)=1$; disjoint $A_i \Rightarrow P(\cup A_i) = \sum P(A_i)$
$P(A\cup B) = P(A) + P(B) - P(A\cap B)$
$P(A^c) = 1 - P(A)$; $P(\emptyset) = 0$; $A \subset B \Rightarrow P(A) \leq P(B)$
De Morgan: $(A\cup B)^c = A^c \cap B^c$; $(A\cap B)^c = A^c \cup B^c$

$$P(A \mid B) = \frac{P(A\cap B)}{P(B)}$$

Bayes: $P(A \mid B) = \dfrac{P(B \mid A)P(A)}{P(B)}$, with $P(B) = \sum_i P(B \mid A_i)P(A_i)$ over a partition $\{A_i\}$
Independent events: $P(A\cap B) = P(A)P(B)$

12. CDF FACTS

$F_X(x) = P(X \leq x)$: non-decreasing, right-continuous, $F(-\infty)=0$, $F(\infty)=1$

$$f_X(x) = \frac{dF_X(x)}{dx} \text{ for continuous } X$$

$P(a < X \leq b) = F(b) - F(a)$
A pdf may exceed 1; a pmf may not. What must equal 1 is the total integral or sum.

13. EXAM PROCEDURE REMINDERS

- Say "by independence" wherever a variance of a sum splits.
- Say "identically distributed" wherever $E(X_i)$ becomes μ.
- Name the theorem: Thm 5.2.6, Thm 5.3.1(c), invariance property of the MLE, Slutsky, delta method, asymptotic properties of the MLE.
- Evaluate every identity at θ_0.
- Give the form requested; when unsure, write both the limiting and the asymptotic distribution.
- Label estimators $\widehat{\theta}_{MLE}$ or $\widehat{\theta}_{MOM}$.
- Answer a yes/no question with "Yes" or "No" first, then justify.
- A labeled setup on an unfinished problem beats a blank.

14. GAMMA FUNCTION AND USEFUL LIMITS

$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1}e^{-t}dt$
$\Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$; $\Gamma(n) = (n-1)!$ for integer $n \geq 1$
$\Gamma(1) = 1$; $\Gamma(1/2) = \sqrt{\pi}$

$$\lim_{n\to\infty}\frac{n-1}{n} = 1, \quad \lim_{n\to\infty}\frac{c}{n} = 0, \quad \lim_{n\to\infty}\frac{c}{n-1} = 0$$
$$\lim_{n\to\infty}\frac{n}{n+1} = 1, \quad \lim_{n\to\infty}\left(1+\tfrac{a}{n}\right)^n = e^a$$

Anything of the form $\frac{\text{constant in } n}{\text{polynomial in } n}$ goes to zero, which is what almost every consistency proof needs.

15. WHERE THINGS LIVE IN THE OTHER SHEET

Pipeline, score, $I(\theta_0)$, CRLB → Sheet 1 box 1
MLE steps and invariance → Sheet 1 box 2
MOM steps → Sheet 1 box 3
Delta method → Sheet 1 box 4
UMVUE test → Sheet 1 box 5
Consistency ritual → Sheet 1 box 6
Proof skeletons → Sheet 1 box 8
χ^2, t, F construction → Sheet 1 box 8 (side B)
Sampling distribution facts → Sheet 1 boxes 9, 9b
MSE → Sheet 1 box 10
WLLN, CLT, Slutsky, MLE theorem → Sheet 1 boxes 11, 11b
Score and information identities → Sheet 1 box 12
Regularity failure → Sheet 1 box 13